Hardware Lab 1

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Part I Half Adder

Generates a Sum and a Carry output

Sum

A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

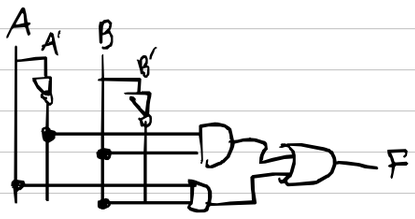
	B	$\bar{B}$
$\bar{A}$	1	0
A	0	1

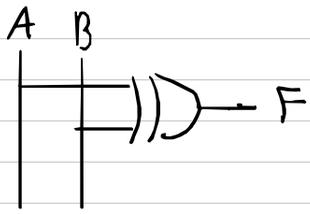
Boolean Expression

$\bar{A}B + A\bar{B}$ or $\overline{AB + \bar{A}\bar{B}}$ → Sum

Digital Logic for Sum

$\bar{A}B + A\bar{B}$





To simplify F use XOR

$A \oplus B$

Because

$\bar{A}B + A\bar{B}$ is resulting in 1/true when A and b are different

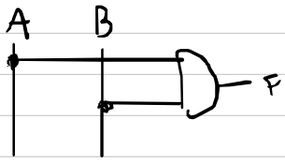
K-map (carry)

	B	$\bar{B}$
$\bar{A}$	0	0
A	1	0

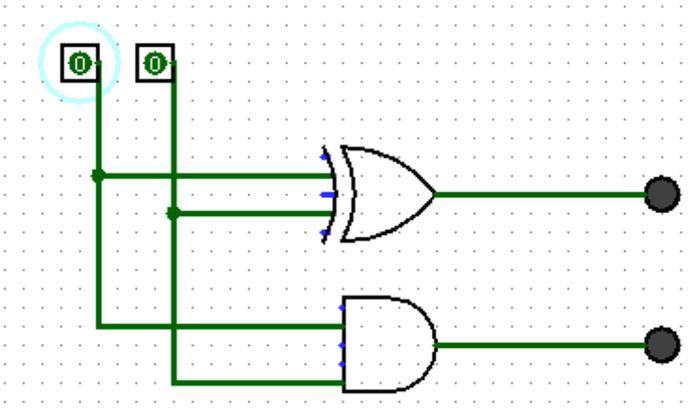
Boolean Expression

AB or $\overline{\bar{A} + \bar{B}}$ → carry

Digital Logic



Combined Digital Logic



Part I Full Adder

Generates a sum and a Carry For 3 variables

Truth Table

A	B	C	Sum	CARRY
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K-map For Sum

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	0	1
A	1	0	1	0

$(A\bar{B}\bar{C}) + (\bar{A}\bar{B}C) + (ABC) + (\bar{A}B\bar{C})$

$A(\bar{B}\bar{C} + BC) + \bar{A}(\bar{B}C + B\bar{C})$

$A(\overline{B \oplus C}) + \bar{A}(B \oplus C)$

$A\bar{X} + \bar{A}X$

$A \oplus X$

$A \oplus B \oplus C$

K-map For CARRY

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	3	2
A	1	5	7	6

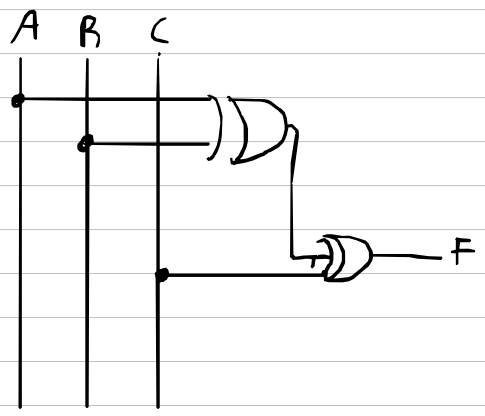
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	0	1	0
A	0	1	1	1

$(AB) + (AC) + (BC)$

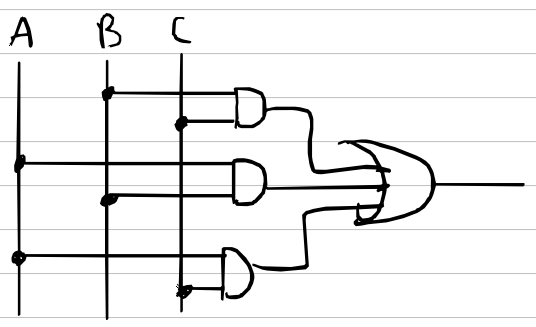
$BC + A(B + C)$

Circuit Diagram

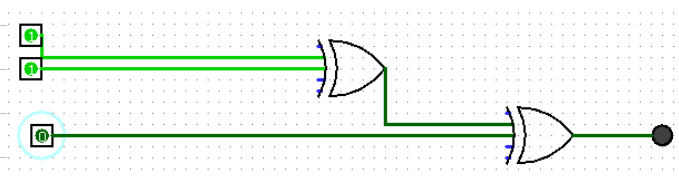
For Sum

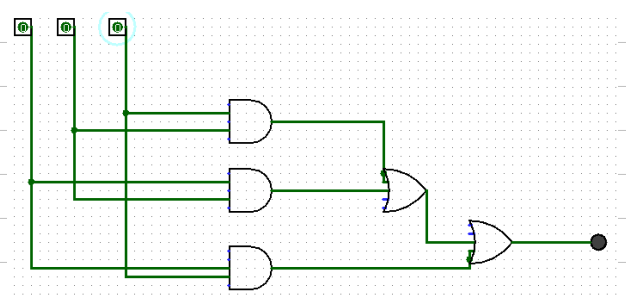


For Carry

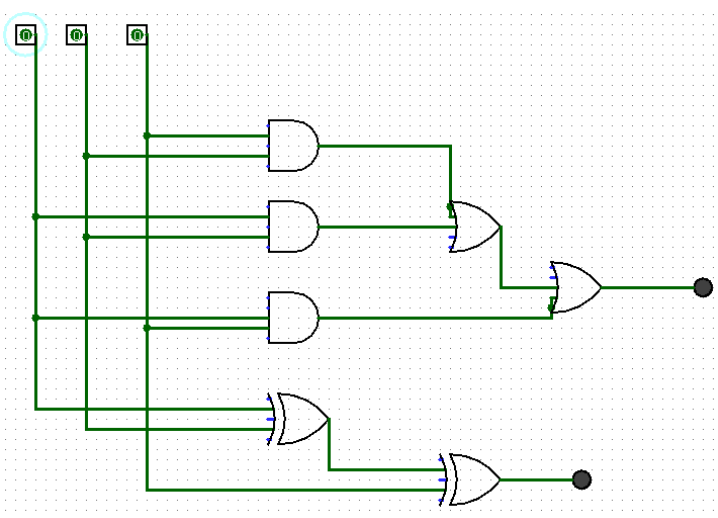


Logisim Models





Combined Circuit -->

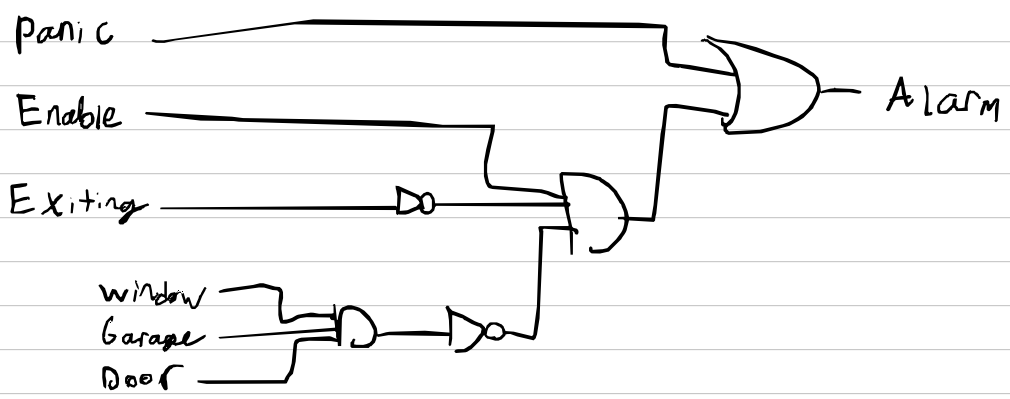


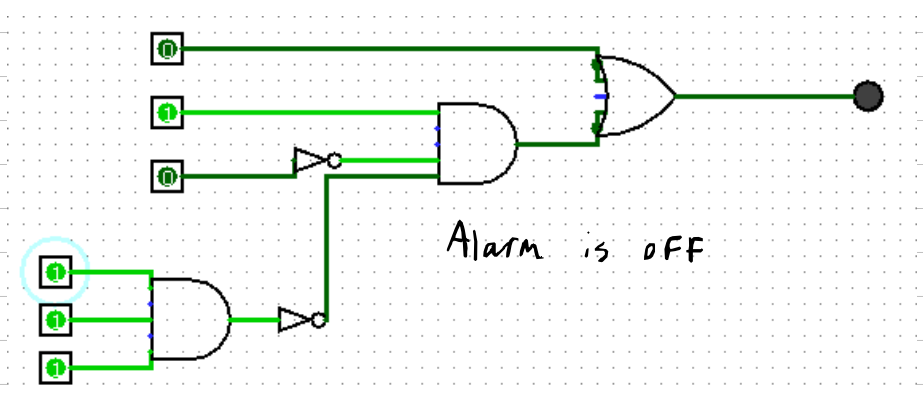
Part II Home Security

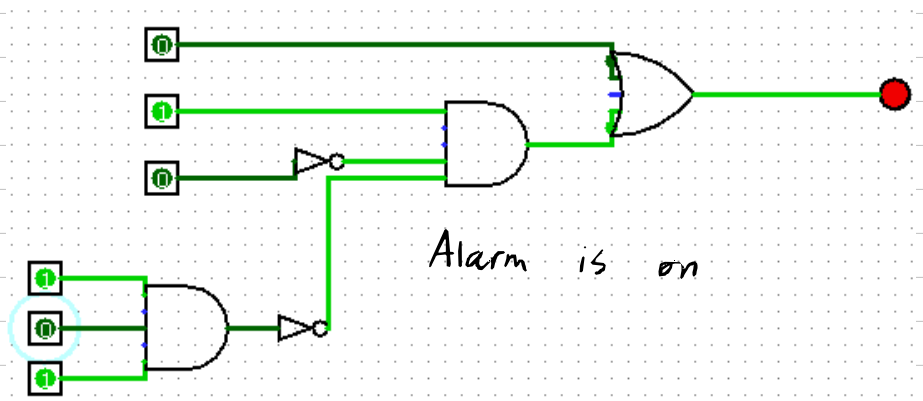
Truth Table

Panic	Enable	Exiting	Window	Door	Garage	Alarm
1	x	x	x	x	x	1
x	1	0	0	0	0	1
x	1	0	1	0	0	1
x	1	0	0	1	0	1
x	1	0	0	0	1	1
x	1	0	1	1	0	1
x	1	0	1	0	1	1
x	1	0	0	1	1	1

Circuit







Part III Difficulties

The hardest part of this lab was by far creating the half-adder circuit. When initally creating the circuit the LEDs would turn on indefinitally. To troubleshoot this problem I first tested my circuit on both Logisim and Tinker CAD. After verifying the circuit was logically sound, I tested each gate by replacing them with identical gates. This test did not yield any changes. After changing the circuit in a number of ways I determined that voltage must be the culprit. Using Tinker CAD I tested the circuit at 10 volts (Which is what the given PSU supplies), this showed that 10 volts was too high for the IC to work properly. After looking at each gates datasheet I found that the IC's can only take a maximum of 7 Volts. To get this circuit to work I implemented an Arduno Nano which supplied 5 Volts. Once connecting the circuit to a 5 Volt input the circuit worked.